Joo-Kyung Kim

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Experience

Amazon Bellevue, WA
Senior Applied Scientist *Aug 2017 – Present*

- Building core LLM components powering Alexa for hundreds of millions of users. Specializing in LLM post-training for multi-turn/proactive interaction, agentic RAG, personalization, clarification/disambiguation, and reward modeling.
- Founded and tech-led LLM post-training for multi-turn/proactive interaction (2024–):
  - Developed LLM post-training workflow and toolkit for multi-turn/agentic reinforcement learning.
  - Increased engagement by 70%, ads impressions by 70%, and shopping impressions by 170%, while reducing latency by 60% compared to the prior LLM.
- Designed and tech-led an LLM-based open-ended conversational recommendation system (2024–2026):
  - Deployed for follow-up suggestions/questions/ads; 3x engaged users and 2x engagement rate over a base LLM.
- Designed and tech-led conversation summarization/retrieval system (2024):
  - Enabled cross-device conversation continuity for Alexa.
- Founded and tech-led disambiguation/clarification modeling across multi-turn interactions (2020–2023):
  - Reduced millions of customer-facing defects per week via disambiguation/clarification.
- Drove hypothesis ranking for large-scale natural language understanding (2017–2020):
  - Shipped production models enabling dynamic routing across thousands of first- and third-party Alexa skills.

Microsoft Bellevue, WA
Research Intern *May 2016 – Aug 2016*

- Deep learning on conversational understanding (Mentors: Gokhan Tur, Asli Celikyilmaz)

NEC Laboratories Princeton, NJ
Research Associate *May 2015 – Aug 2015*

- Constituency parsing with deep and recursive neural networks (Mentor: Bing Bai)

Nuance Sunnyvale, CA
Research Intern *Jun 2014 – Aug 2014*

- Named entity recognition with neural networks for spoken dialog systems (Mentor: Adwait Ratnaparkhi)

Naver Seongnam, Korea
Software Engineer *Sep 2008 – Jul 2010*

- Web search modeling for Naver Japan, and developed integrated Naver search services

Education

The Ohio State University Columbus, OH
Ph.D., Department of Computer Science and Engineering *Aug 2017*

- Dissertation: Linguistic Knowledge Transfer for Enriching Vector Representations
- Advisor: Eric Fosler-Lussier

Seoul National University Seoul, Korea
M.S., School of Computer Science and Engineering *Aug 2008*

- Thesis: Evolutionary optimization of a collection of variable-length subpatterns for pattern classification
- Advisor: Byoung-Tak Zhang

Sogang University Seoul, Korea
B.E., Department of Computer Science *Aug 2005*

- Thesis: Real-time news categorization system
- Advisor: Jihoon Yang

Publications

Jinyoung Park, Sanghyeok Lee, Omar Zia Khan, Hyunwoo J. Kim, and **Joo-Kyung Kim**. "HyperGraphPro: Progress-Aware Reinforcement Learning for Structure-Guided Hypergraph RAG". In: *NeurIPS 2026 Workshop on Transitioning from Pre-Training to Post-Training (Pre-to-Post)*. 2026.

Xinyi Liu, Hooshang Nayyeri, Dilek Hakkani-Tur, Emine Yilmaz, **Joo-Kyung Kim**, Yifei Zhang, Charith Peris, and Hari Thadakamalla. "Hear2Act: Benchmarking When Prosody Should Change What an Assistant Does". In: *NeurIPS 2026 Workshop on Real-Time Conversational Agents (RTCA)*. 2026.

Soyeong Jeong, Taehee Jung, Sung Ju Hwang, **Joo-Kyung Kim**, and Dongyeop Kang. "When Thoughts Meet Facts: Reusable Reasoning for Long-Context LMs". In: *ACL Findings*. 2026.

Qiusi Zhan, Angeline Budiman-Chan, Abdelrahman Zayed, Xingzhi Guo, Daniel Kang, and **Joo-Kyung Kim**. "SafeSearch: Do Not Trade Safety for Utility in LLM Search Agents". In: *EACL Findings*. 2026.

Zae Myung Kim, Anand Ramachandran, Farideh Tavazoee, **Joo-Kyung Kim**, Oleg Rokhlenko, and Dongyeop Kang. "Align to Structure: Aligning Large Language Models with Structural Information". In: *AAAI*. 2026.

Chanwoo Park, Seungju Han, Xingzhi Guo, Asuman Ozdaglar, Kaiqing Zhang, and **Joo-Kyung Kim**. "MAPoRL: Multi-Agent Post-Co-Training for Collaborative Large Language Models with Reinforcement Learning". In: *ACL*. 2025.

Shirley Anugrah Hayati, Taehee Jung, Tristan Bodding-Long, Sudipta Kar, Abhinav Sethy, **Joo-Kyung Kim**, and Dongyeop Kang. "Chain-of-Instructions: Compositional Instruction Tuning on Large Language Models". In: *AAAI*. 2025.

Jinyoung Park, Minseok Joo, **Joo-Kyung Kim**, and Hyunwoo J. Kim. "Generative Subgraph Retrieval for Knowledge Graph–Grounded Dialog Generation". In: *EMNLP*. 2024.

Jihyung Kil, Farideh Tavazoee, Dongyeop Kang, and **Joo-Kyung Kim**. "II-MMR: Identifying and Improving Multi-modal Multi-hop Reasoning in Visual Question Answering". In: *ACL Findings*. 2024.

Jinyoung Park, Ameen Patel, Omar Zia Khan, Hyunwoo J. Kim, and **Joo-Kyung Kim**. *Graph Elicitation for Guiding Multi-Step Reasoning in Large Language Models*. 2023. arXiv: 2311.09762.

Taehee Jung, **Joo-Kyung Kim**, Sungjin Lee, and Dongyeop Kang. "Cluster-Guided Label Generation in Extreme Multi-Label Classification". In: *EACL*. 2023.

Joo-Kyung Kim, Guoyin Wang, Sungjin Lee, and Young-Bum Kim. "Deciding Whether to Ask Clarifying Questions in Large-Scale Spoken Language Understanding". In: *ASRU*. 2021.

Joo-Kyung Kim and Young-Bum Kim. "Pseudo Labeling and Negative Feedback Learning for Large-scale Multi-label Domain Classification". In: *ICASSP*. 2020.

Joo-Kyung Kim and Young-Bum Kim. "Supervised Domain Enablement Attention for Personalized Domain Classification". In: *EMNLP*. 2018.

Joo-Kyung Kim and Young-Bum Kim. "Joint Learning of Domain Classification and Out-of-Domain Detection with Dynamic Class Weighting for Satisficing False Acceptance Rates". In: *Interspeech*. 2018.

Young-Bum Kim, Dongchan Kim, **Joo-Kyung Kim**, and Ruhi Sarikaya. "A Scalable Neural Shortlisting-Reranking Approach for Large-Scale Domain Classification in Natural Language Understanding". In: *NAACL*. 2018.

Joo-Kyung Kim, Young-Bum Kim, Ruhi Sarikaya, and Eric Fosler-Lussier. "Cross-Lingual Transfer Learning for POS Tagging without Cross-Lingual Resources". In: *EMNLP*. 2017.

Joo-Kyung Kim, Gokhan Tur, Asli Celikyilmaz, Bin Cao, and Ye-Yi Wang. "Intent Detection using Semantically Enriched Word Embeddings". In: *SLT*. 2016.

Joo-Kyung Kim, Marie-Catherine de Marneffe, and Eric Fosler-Lussier. "Adjusting Word Embeddings with Semantic Intensity Orders". In: *ACL Workshop on Representation Learning for NLP (RepL4NLP)*. 2016.

Joo-Kyung Kim, Marie-Catherine de Marneffe, and Eric Fosler-Lussier. "Neural word embeddings with multiplicative feature interactions for tensor-based compositions". In: *NAACL Workshop on Vector Space Modeling for NP (VSM)*. 2015.

Joo-Kyung Kim and Marie-Catherine de Marneffe. "Deriving adjectival scales from continuous space word representations". In: *EMNLP*. 2013.

Joo-Kyung Kim and Byoung-Tak Zhang. "Evolving hypernetworks for pattern classification". In: *IEEE Congress on Evolutionary Computation (CEC)*. 2007.

Joo-Kyung Kim, Byung Soo Kim, Oh Kyuk Kwon, Seung Kon Hwang, Jung-Woo Ha, Chan-Hoon Park, Duck Jin Chung, Chong Ho Lee, Jaehyun Park, and Byoung-Tak Zhang. "A DNA computing-inspired silicon chip for pattern recognition". In: *International Meeting on DNA Computing (DNA)*. 2007.

Byoung-Tak Zhang and **Joo-Kyung Kim**. "DNA hypernetworks for information storage and retrieval". In: *International Meeting on DNA Computing (DNA)*. 2006.

Academic Service

Senior Area Chair

- ARR 2025–2026, EMNLP 2025–2026, ACL 2026

Area Chair

- ARR Dec 2023 – Feb 2025

Reviewer

- ACL 2019–2023, EMNLP 2019–2023, NAACL 2019–2022, EACL 2021–2023, AACL 2023, AAAI 2020–2022, IEEE TASLP 2020–2021

Intern Mentorship

2025 Soyeong Jeong (KAIST) – *Template Retrieval for Long-Context LLMs*

2025 Qiusi Zhan (UIUC) – *Safety Alignment in Agentic RAG*

2024 Chanwoo Park (MIT) – *Multi-Agent Reinforcement Learning for LLMs*

2024 Zae Myung Kim (UMN) – *Structural LLM Alignment for Long-form Generation*

2024 Shirley Anugrah Hayati (UMN) – *Diverse and Distribution-Controlled Generation with LLMs*

2023 Shirley Anugrah Hayati (UMN) – *Compositional Instruction Tuning on LLMs*

2023 Jihyung Kil (OSU) – *Multi-modal Multi-hop Reasoning in VQA*

2023 Jinyoung Park (Korea University) – *Multi-Step Reasoning in LLMs*

2022 Jinyoung Park (Korea University) – *Generative Retrieval for Grounded Dialog Generation*

2021 Taehee Jung (University of Pittsburgh) – *Generative Extreme Multi-Label Classification*

Skills

Programming Languages

- Python, Java, C/C++, Lua, Objective C, Delphi, PHP